Standardization of Pacific Blue Marlin Catch Per Unit Effort in the Hawaii Longline Fishery from 1995-2024

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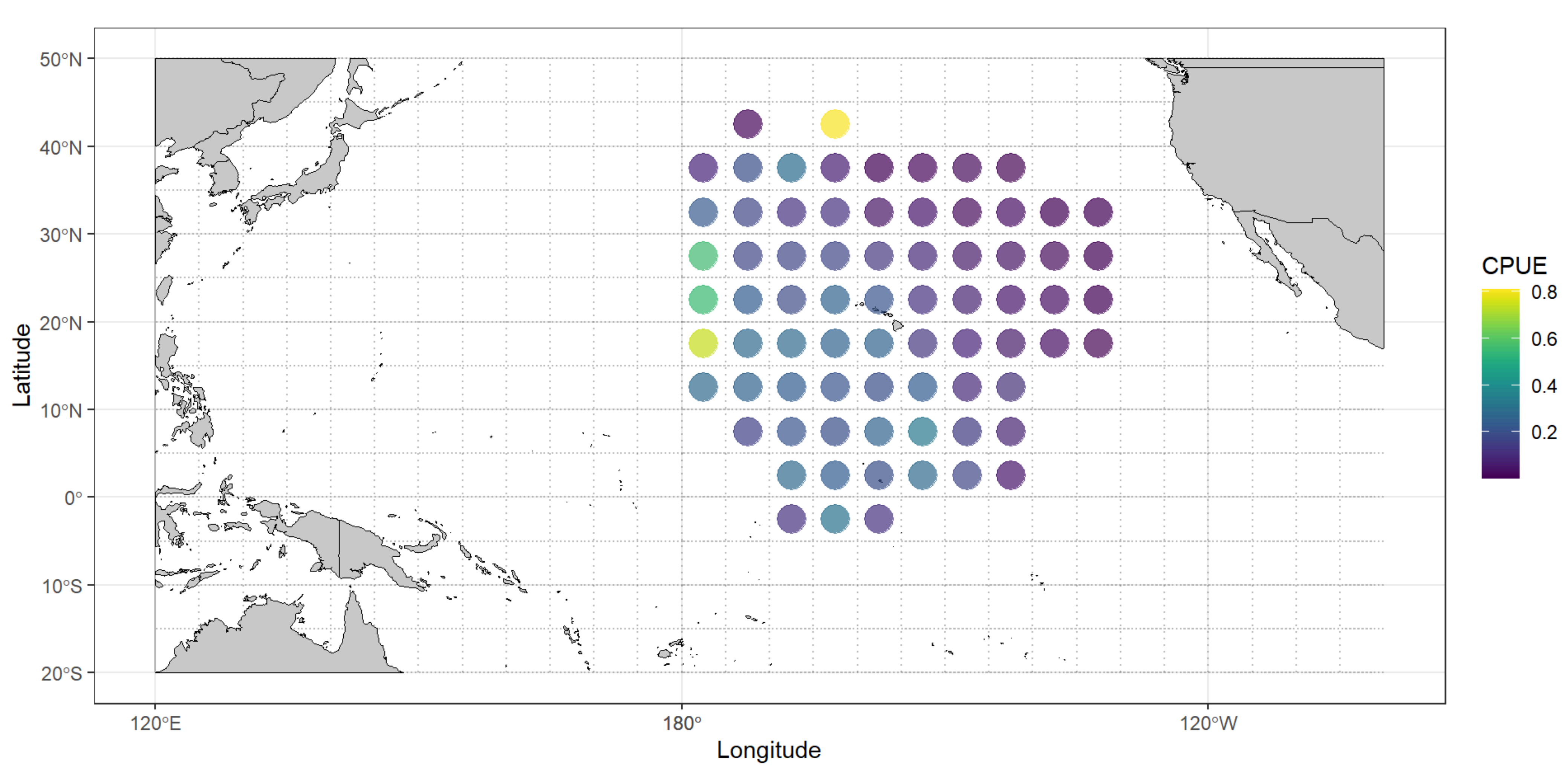
Pacific blue marlin in the -

- Indo-Pacific blue marlin are caught by several fleets in the US:
  - Hawaii longline (2 sectors targeting swordfish and tuna, respectively)
  - American Samoa longline
  - Handline
  - Troll
- Both HI LL and AS LL have observer program with varying amounts of coverage.
- BUM is a non-targeted but economically important species in the LL fisheries
- Overall catch appears to have increased in the last 10 years

Data

- Data from 1995 - 2024 from the Pacific Islands Longline Logbook Program
- Deep Set :
  - Targets tuna
  - ≥ 10 hooks per float 1994-2003
  - ≥15 hooks per float after 2004
  - 3-10% observer coverage prior to 2004, 20% 2004 - 2020. 20-7% since 2020
  - accounts for 90% of the catch
- Shallow Set:
  - Targets Swordfish
  - <10 hooks per float 1994-2000
  - <15 hooks per float after 2004
  - Fishery closed from 2001-2004
  - 100% observer coverage after 2004

Blue Marlin Nominal CPUE

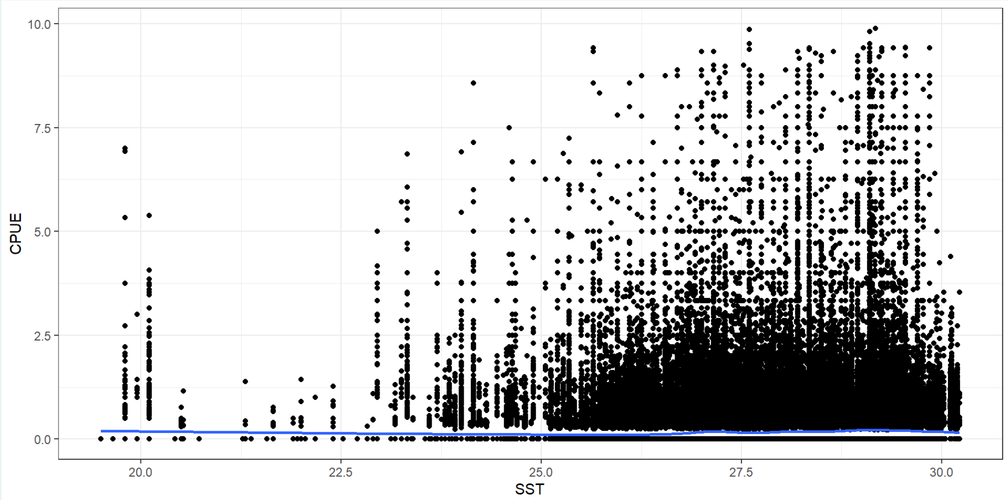


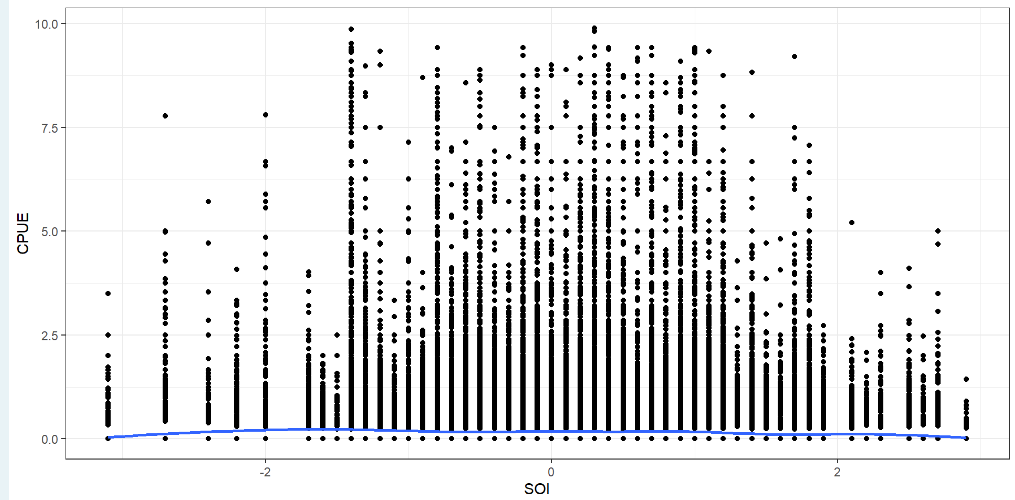
Model Structure

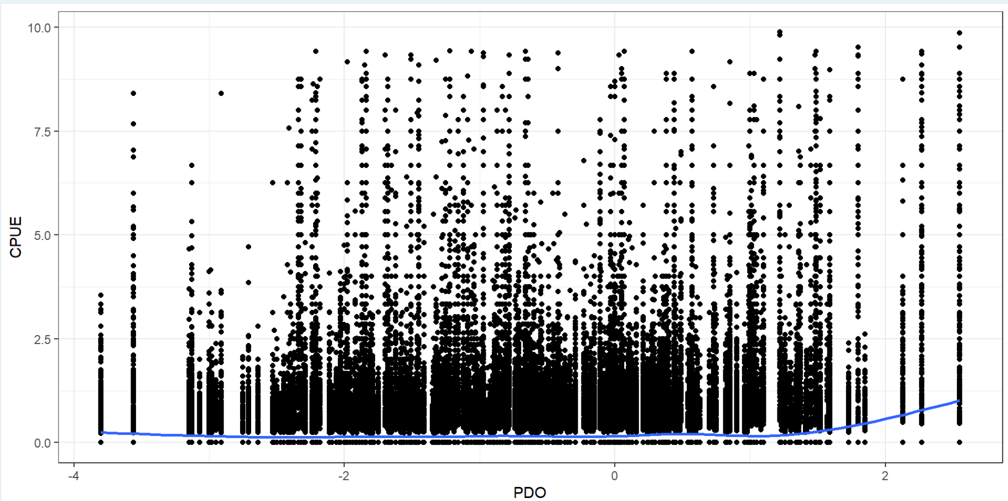
- Delta-lognormal GLMM for both fleets combined
  - lognormal distribution for positive catches
  - binomial distribution with logit-link function for encounter probability
- Forward step-wise model selection on 13 potential explaintory variables
- Inclusion criteria were based upon an increase of percent deviance explained at least 0.10% and a Chi-squared likelihood ratio test

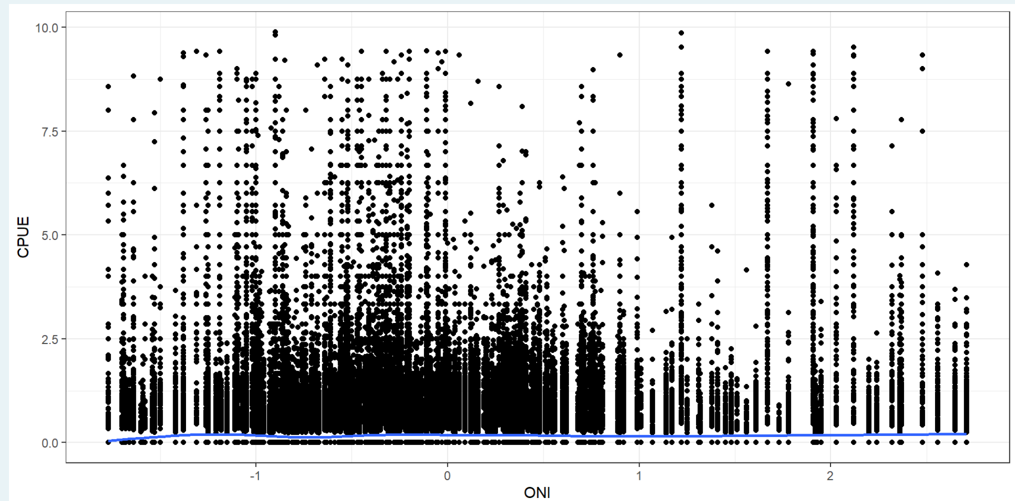
Model Structure

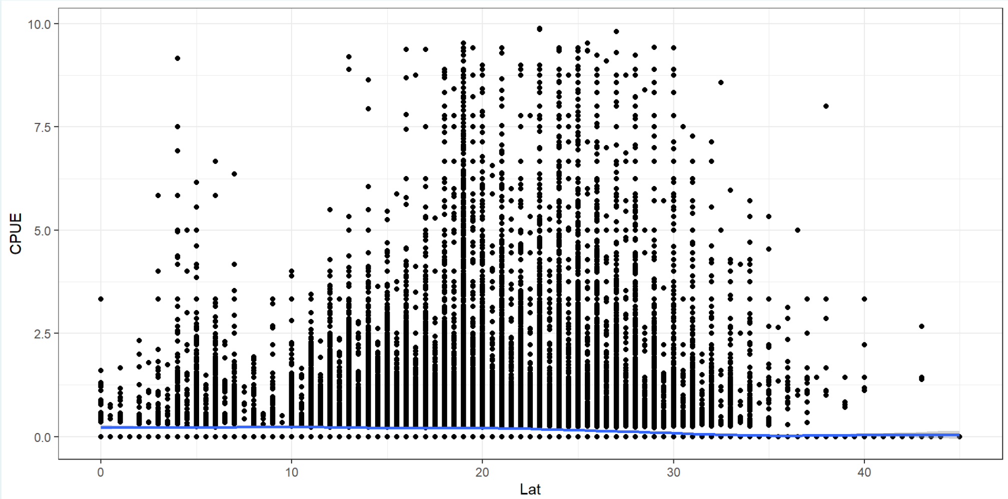
- Operational variables:
  - Year
  - Month
  - Quarter
  - Hooks per float
  - bait type
  - Set time (Continuous and factor)
  - Set type
  - Vessel (random effect)
- Spatial and environmental
  - Latitude
  - Longitude
  - SST
  - PDO
  - SOI
  - ONI

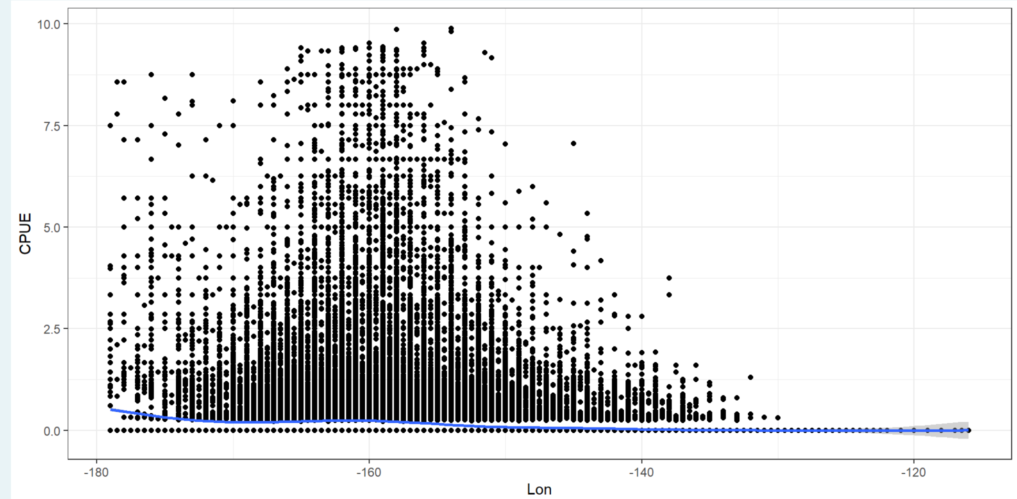

CPUE
SST
10.0
7.5
5.0
2.5
0.0
20.0
22.5
25.0
27.5
30.0

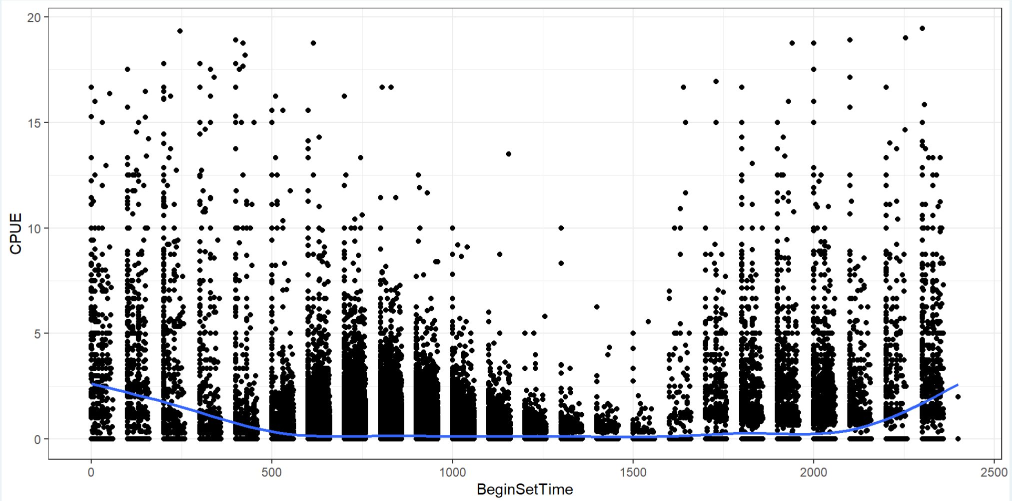

CPUE
SOI
10.0
7.5
5.0
2.5
0.0
-2
0
2

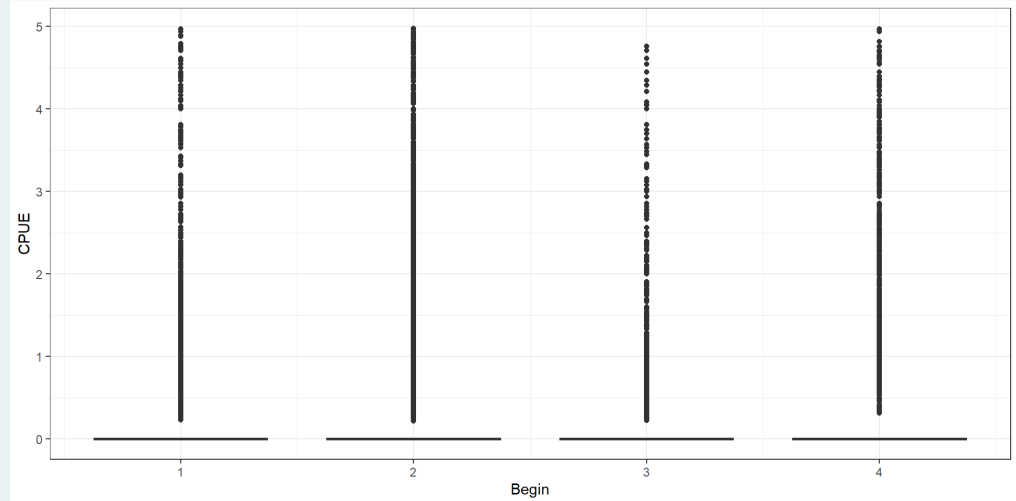

CPUE
PDO
10.0
7.5
5.0
2.5
0.0
-4
-2
0
2

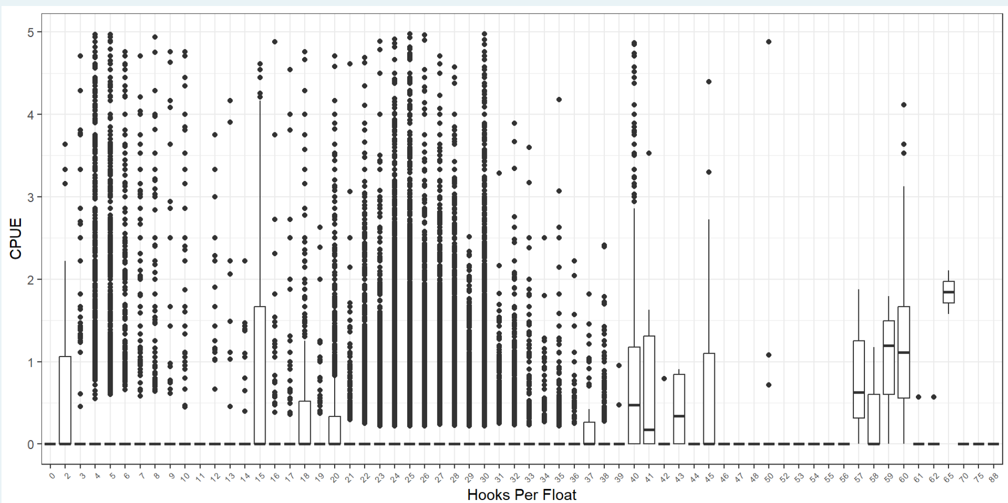

CPUE
ONI
10.0
7.5
5.0
2.5
0.0
-1
0
1
2

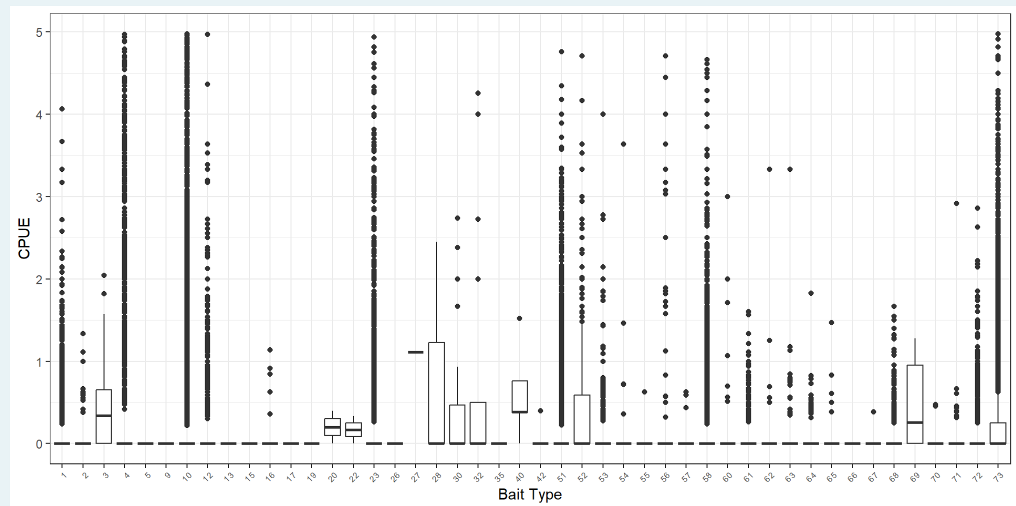

CPUE
Lat
10.0
7.5
5.0
2.5
0.0
0
10
20
30
40


CPUE
Lon
10.0
7.5
5.0
2.5
0.0
-180
-160
-140
-120


CPUE
BeginSetTime
0
500
1000
1500
2000
2500
5
10
15
20


CPUE
Begin
1
2
3
4


CPUE
Hooks Per Float


CPUE
Bait Type

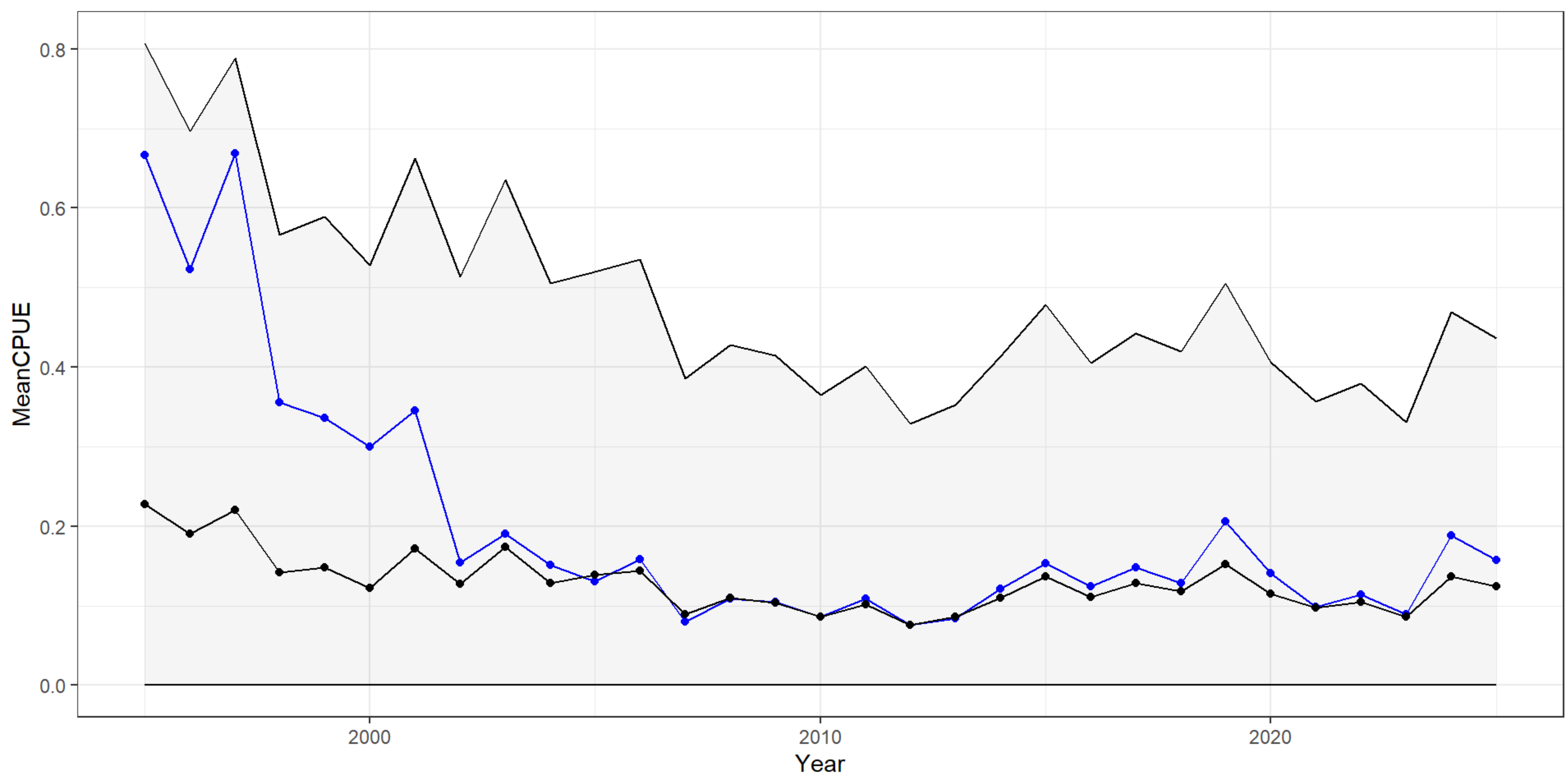
Final Models

$$Probability of Encounter \sim Latitude + Year + Month + Longitude + Bait$$

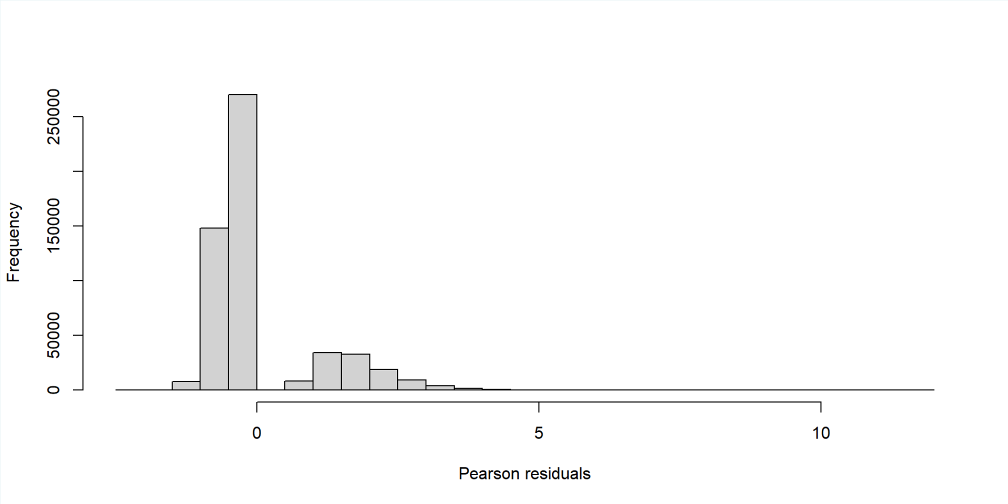
9.7% Deviance Explained

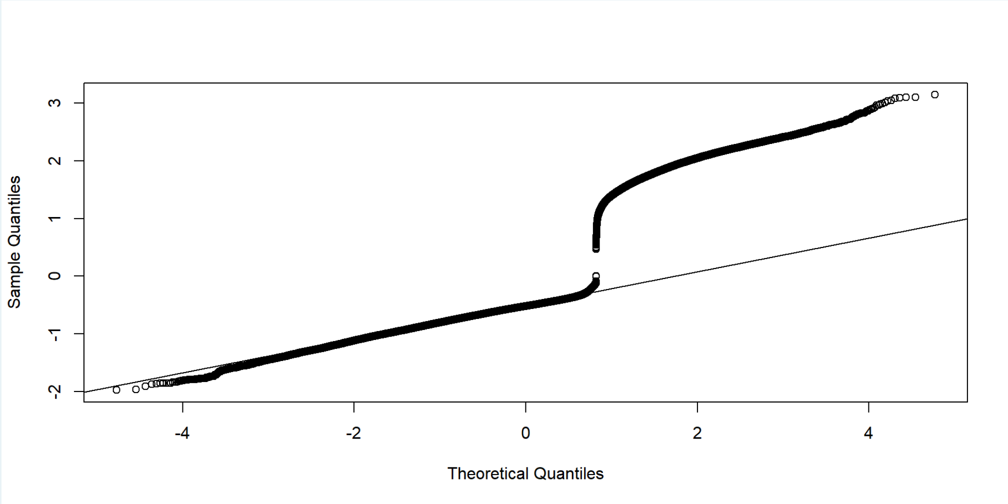
$$log(CPUE) \sim Bait + Begin + Year + HPF + (1|Vessel)$$

10.2% Deviance Explained

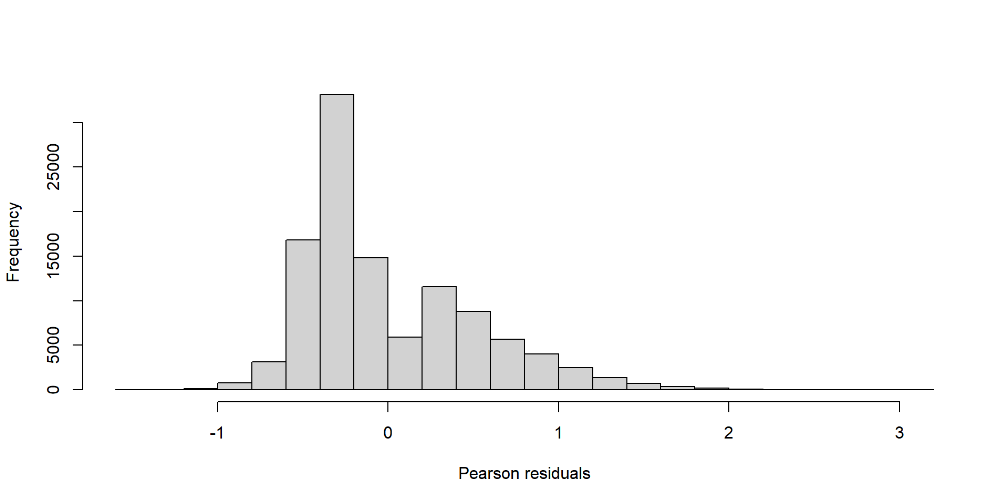

0.8
0.6
0.4
0.2
0.0
MeanCPUE
2000
2010
2020
Year

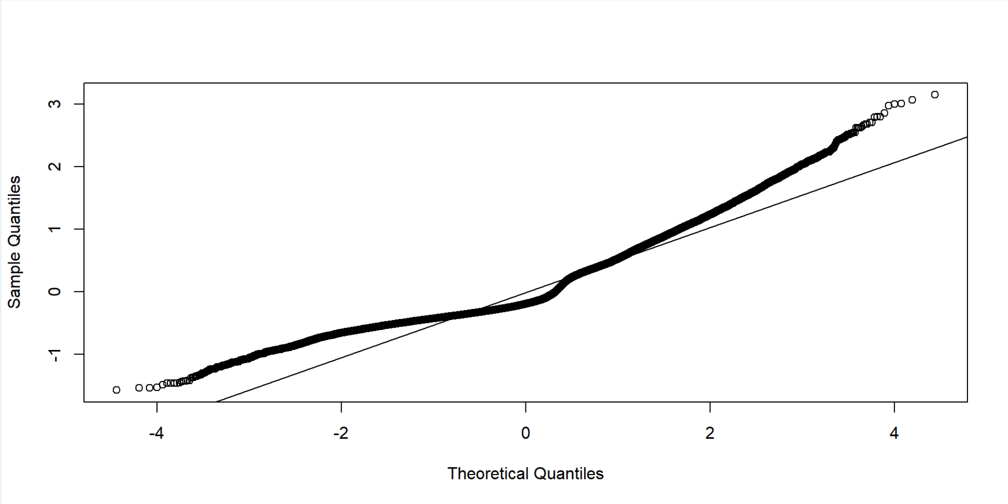
Diagnostics - Encounter Probability



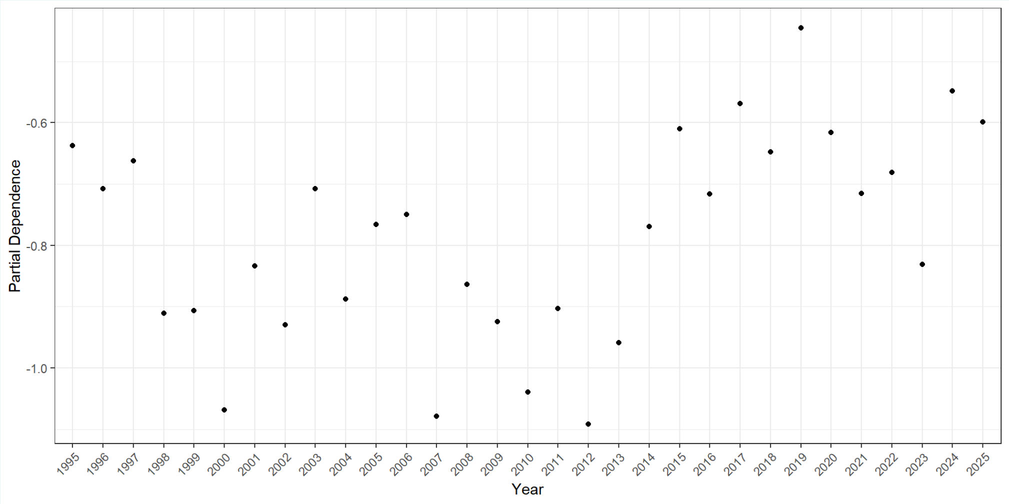


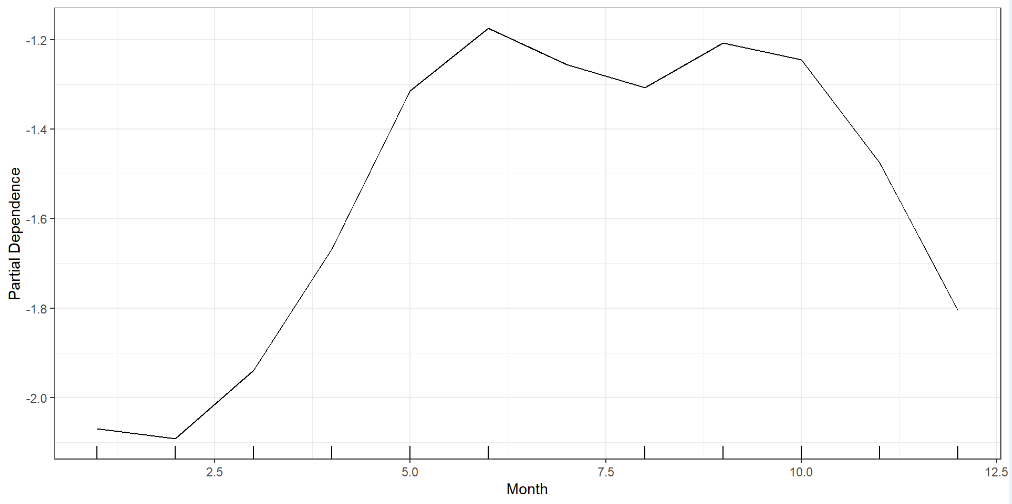
Diagnostics - Positive Catches

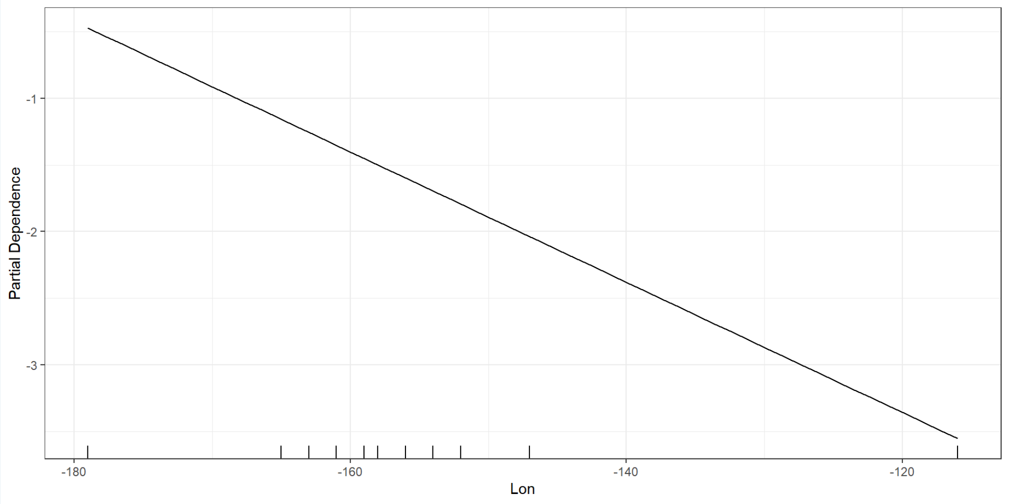


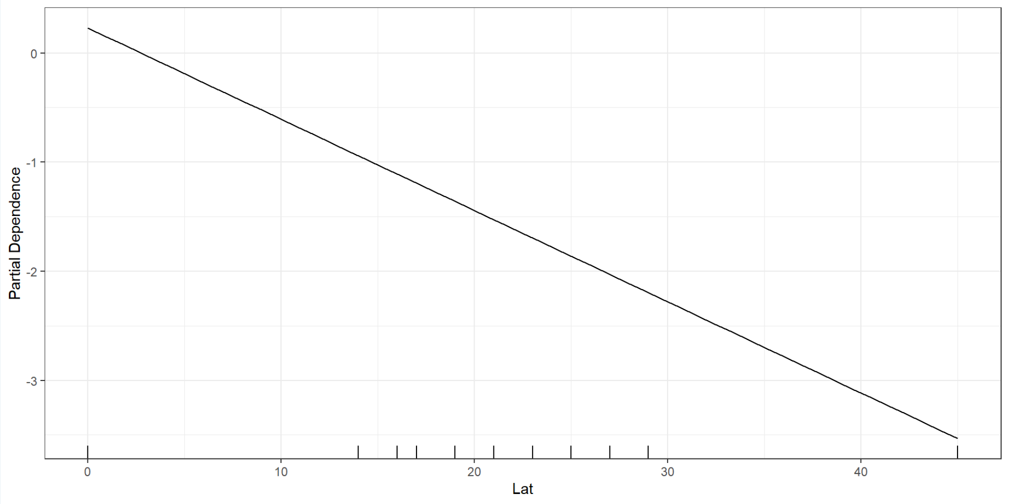


Partial Dependency Plots

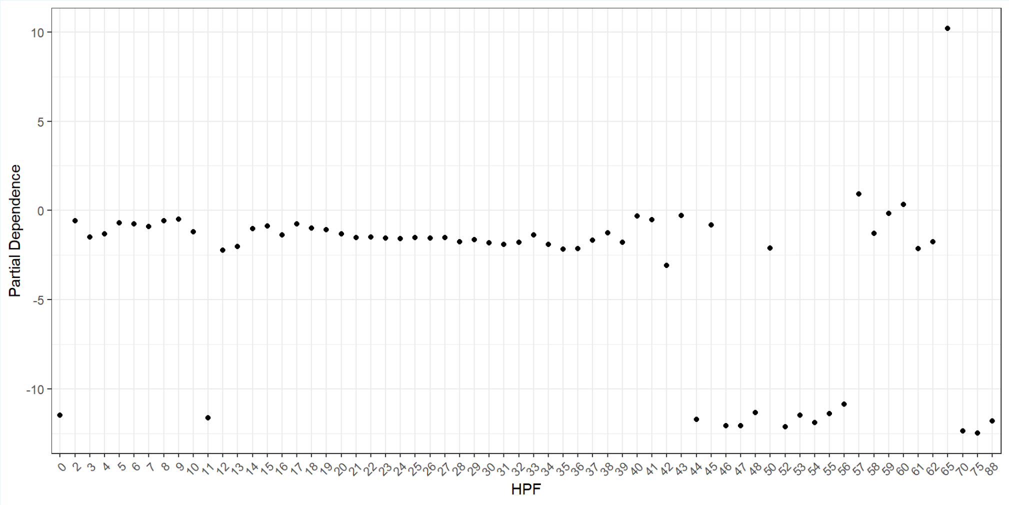


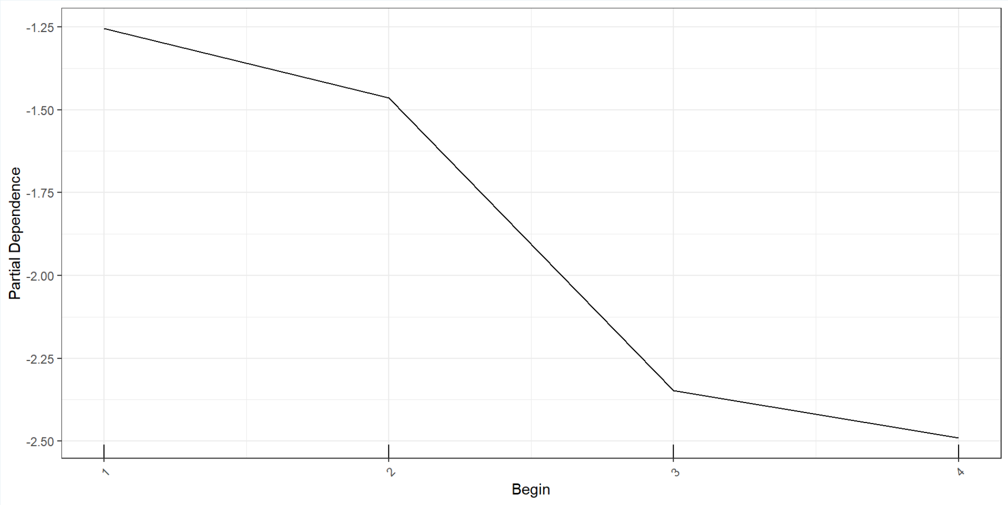


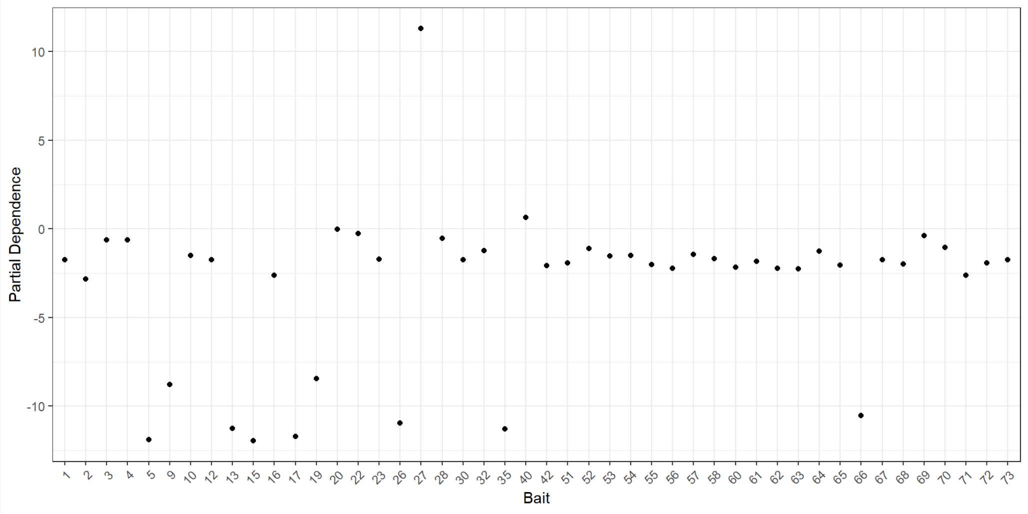




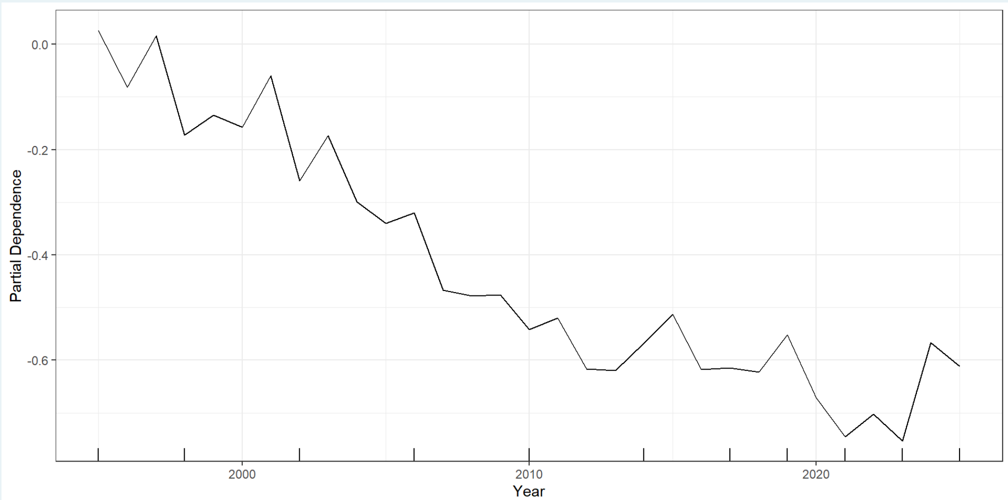
Partial Dependency Plots

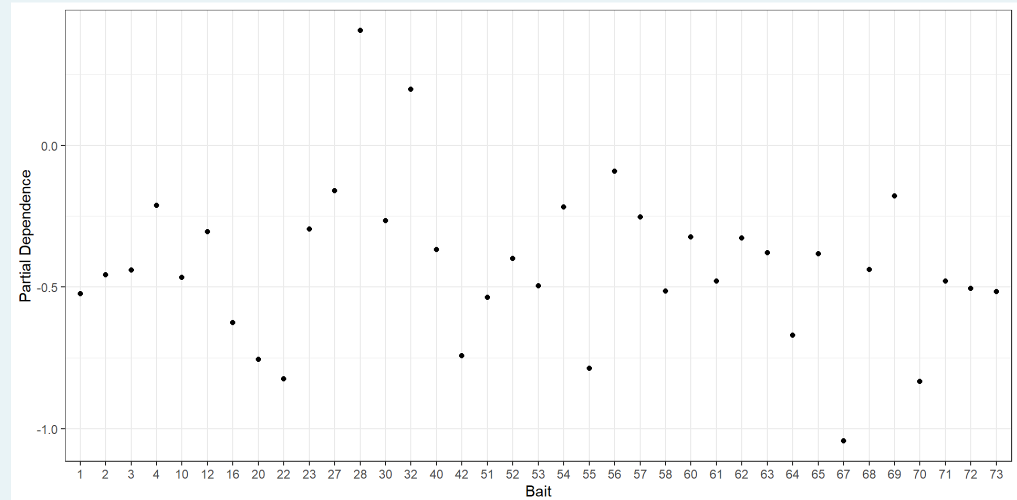


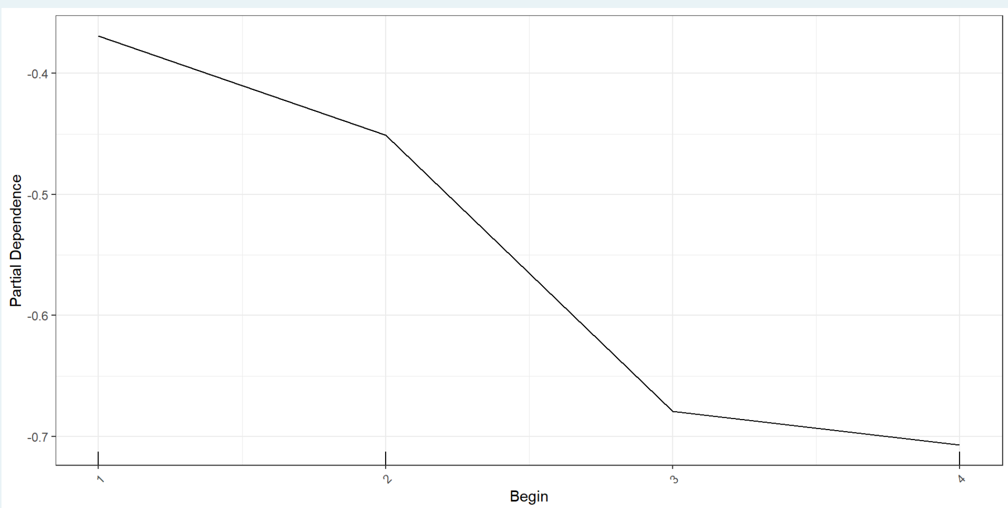


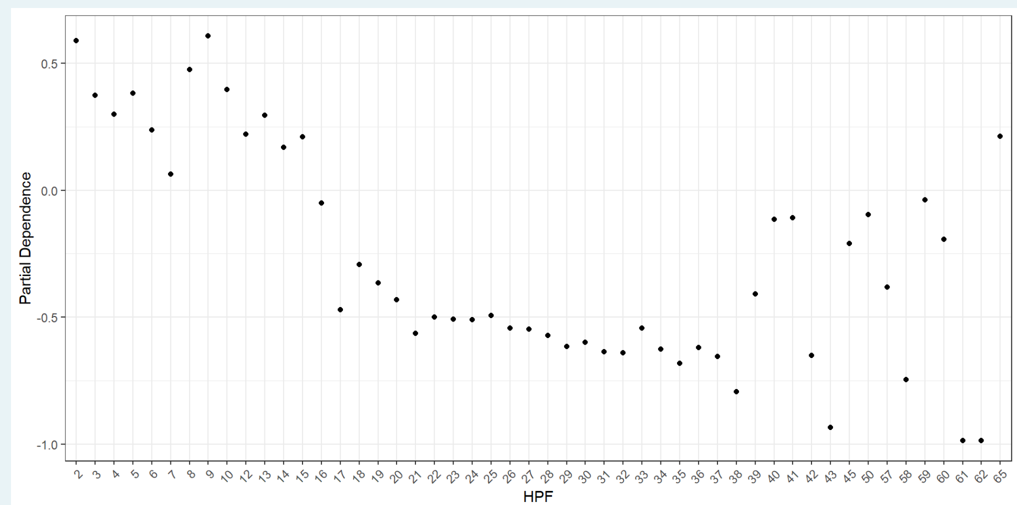


Partial Dependency Plots

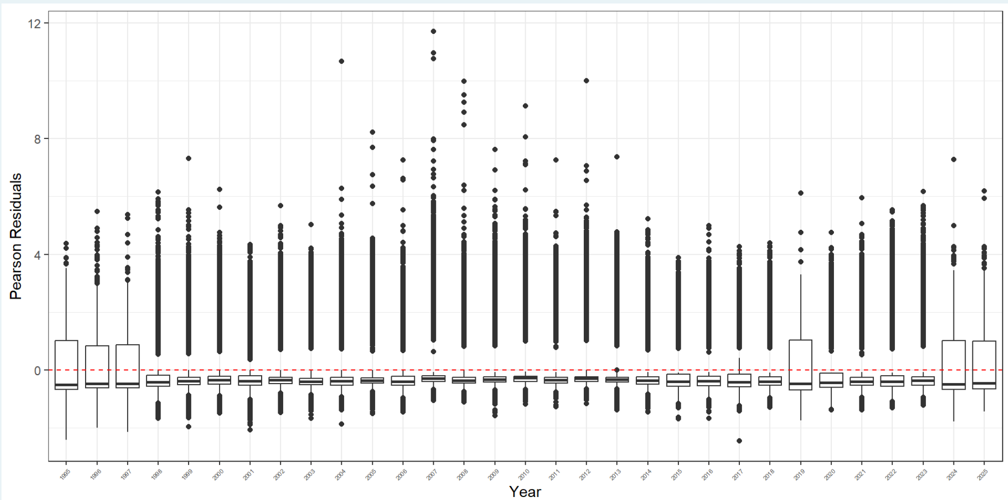


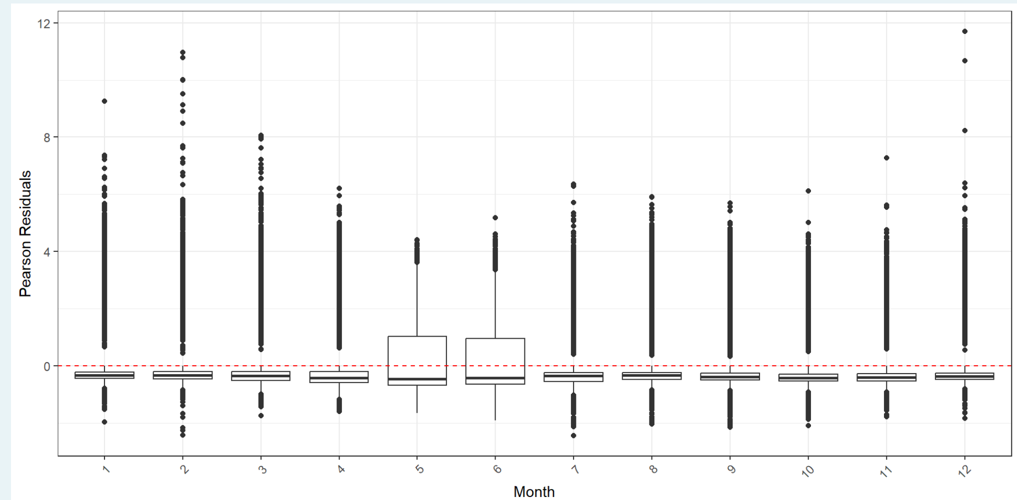




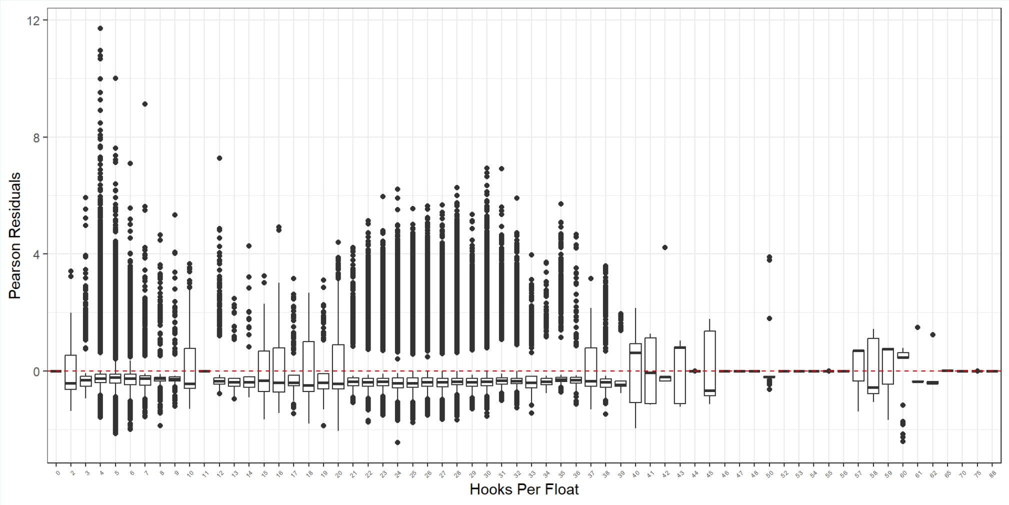


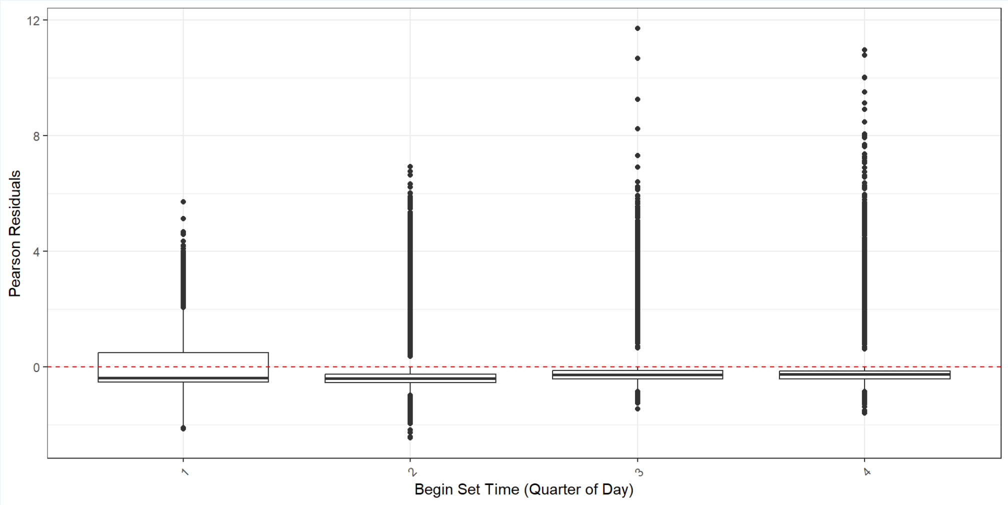
Residual plots - Encounter Probability





Residual plots - Encounter Probability





Residual plots - Positive catches